Morsal Mubarak

Computational Researcher & Software Developer at the Intersection of Data Science, Neuroscience, and Public Health

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EDUCATION

Justice High School, Falls Church, Virginia · Class of 2027 · IB Diploma candidate · IB MYP Certificate — Weighted GPA 4.519 · SAT 1600

RESEARCH

Independent research study · Sole author · Under review, *Scientific Reports* — Aug 2026

- *A Reproducible Public-Data Benchmark of Birthweight-for-Gestational-Age Deviations and Infant/Fetal Mortality Risk: A Cross-Sectional Ecological Analysis of United States Vital Statistics, 2016–2024.*
- Built from **33,589,282** U.S. natality records (NCHS public-use files); code and data are public on GitHub and archived on Zenodo (DOI 10.5281/zenodo.21894038).

Traumatic brain injury (TBI) MRI computational study, Mass General Brigham / MGH Martinos Center · Lead Computational Researcher — Spring 2026

- Built a verified Python pipeline (~4,800 lines; pandas, NumPy, SciPy, statsmodels) linking MR spectroscopy to post-concussion symptoms in a **35-participant** treatment study, with outlier screening, FDR correction, and an automated re-run audit.

PoxHostAtlas · First & corresponding author, with Deon Nguyen (Arizona State University) — 2026

- Manuscript: *PoxHostAtlas: A reproducible cross-study atlas of host gene-expression remodeling during poxvirus infection.* Re-analyzes public poxvirus RNA-sequencing datasets with one common workflow; **1,834** host genes changed reproducibly across four datasets. Code: github.com/mubarsad012/poxhostatlas.

COVID-BRAIN imaging study, Athinoula A. Martinos Center (MGH) · Co-author (2nd author, manuscript) — Oct 2026

- Poster: *Lower thalamic creatine and glutathione associate with cognition and systemic inflammation in neurological post-acute sequelae of COVID-19 infection from the multicenter COVID-BRAIN imaging cohort.* Clinical Research Day, Mass General Brigham, Oct 2026.

Neurocysticercosis systematic review (ESM-2 + AlphaFold) · Senior (last) author — Manuscript in preparation

2026 Student Research Institute, Harvard Undergraduate OpenBio Laboratory · Mentee — Jun 8 – Jul 31, 2026

Bay Area Medical Explorers · Researcher — 2026

- Co-first author: *Assessing the Associations Between Antiretroviral Treatment Resource Shortage and HIV Prevalence Among Teenagers Residing in Zambia* (Mar 2026).
- First author (of 7): *Academic Pressure and Mental Health in South Asian American High School Students* (Jun 2026).

SOFTWARE

ProxyHealth AI · proxyhealth-ai.onrender.com — Public beta, Oct 2026

- Explains official papers, such as medical bills and benefits letters, in 27 languages: what each paper is, what it asks for, and a step-by-step plan, with every date and amount quoted from the paper and checked in code.
- Python/FastAPI backend; TypeScript/Expo front end (web, iOS, Android). Code: github.com/mubarsad012/ProxyHealth-AI.

LEADERSHIP & COMMUNITY

- **Emergency Afghan evacuation fundraiser** (2021): organized with father and brother; raised **$220,000+**.
- **Medical & legal interpreter** (English–Pashto): one of the main fluent English speakers in my household; interpret for family and refugee community members.
- **Public Health Youth Ambassador**, Fairfax County Health Department: Honors Graduate, Liaison Program; **Mentorship Chair**, Alumni Liaison Program Mentorship & Student Development Committee.
- **Founder & President**, Chemistry Club · **President**, IB Achievers Alliance, GEMS, FBLA, HOSA, Math Honor Society, Science Honor Society · **Co-President**, Al Shami Ethnic Club.
- **Senior Class Representative**, SGA · **Principal Student Advisor**, Junior Class · **Marketing Director**, GoMedQuest · **Editor**, Students Demand Action · **Event Coordinator**, Dental Club · **BigFuture Ambassador**, College Board · SOLE Tutor · Chemistry Teacher Assistant.

- **Programs:** Scientific Enrichment Month, Columbia Univ. Herbert Irving Comprehensive Cancer Center (presented on secondhand smoke and vaping to 1,200+ people) · Virginia Virtual Internship (WBLA), team leader, member-retention strategy for the Northern Virginia Black Chamber of Commerce · Neuroscience & genetics internship on lucid dreaming · Bay Area Medical Explorers, Intern Leader (HIV and mental-health case studies).
- **Member:** Key Club, Leo Club, Self Care Club, Medical Club, National English Honor Society.

HONORS

- **OpenBio Student Research Institute Scholar**, top 2.5% of ~3,000 (International, Gr. 12) · **JYEM / NY-CSEF International Research Competition Finalist**, of 1,425, fetal-growth study (Gr. 12).
- **Virginia Governor's School Finalist**, Health & Medicine (Gr. 11) · **2nd Place** of 300+, Fairfax County Regional Science & Engineering Fair (Gr. 11) · **#1 at school**, FBLA Virtual Personal Finance.
- **National Recognition Program**, School Recognition Award · **Morehouse School of Medicine Award** (Health Worker program) · **Editor-at-Large**, Journalism Program Award · **Index Editor**, Yearbook Award.
- **Certifications:** NIH Good Clinical Practice for Social & Behavioral Research · CITI Program (George Mason University) · Stop the Bleed (Fairfax County Health Department) · Learning Path Public Health Series (2025).
- **2nd place (solo)** of 17 finalists, 2025 PHYAP Community Health Competition final (Georgetown University School of Medicine faculty judges, Gr. 10), for *Nourish to Flourish: Empowering Communities through Food Education and EBT Access*, a policy proposal.

SKILLS

Programming: Python, TypeScript · **Tools:** Git, GitHub, Streamlit, LaTeX (Overleaf) · **Languages:** English, Pashto